

ATHMEEYA M KASHYAP

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EDUCATION

IIIT Hyderabad

2024 - 2029 (Expected)

B.Tech Computer Science + MS Computational Natural Sciences

Hyderabad, India

- Coursework: Data Structures & Algorithms, Design & Analysis of Software Systems, Machine & Data Learning, Discrete Structures, Linear Algebra, Probability & Random Processes.

PROFESSIONAL EXPERIENCE

Software Engineer Intern

Jan 2026 - Apr 2026

SVP India (Social Venture Partners India)

Hyderabad, India

- Delivered SVP India's **partner-engagement platform end to end** and rolled it out to replace the manual spreadsheets the team used to track partners, meetings, and attendance by hand.
- Designed a **12-table** PostgreSQL schema and **60+** REST endpoints behind token auth with **role-based access**, covering partners, meetings, attendance, and analytics with Excel and PDF export.

PROJECTS

ForgeLM: GPT Language Model + CPU Inference Engine 🌀 | C++, AVX2/SIMD, PyTorch, Python

- Trained a GPT-style language model in PyTorch and implemented its **C++ CPU inference engine**: matmuls, attention, KV-cache, int4 quantizer, sampler, and HTTP server.
- Accelerated decoding from **20** to **6,359** tokens/sec (**312x**) with a **KV-cache** and AVX2/FMA SIMD kernels, output staying **bit-identical** to the PyTorch reference.
- Added **int4 quantization (7.1x smaller weights)**, a paged KV-cache serving **16** concurrent requests at **8,362** tokens/sec, and speculative decoding that matches greedy exactly.

AlphaExchange: Distributed Benchmarking Platform 🌀 | Go, gRPC/Protobuf, Kafka, Redis, Docker, Kubernetes

- Architected a **distributed** benchmarking platform of **6** Go microservices where users upload order-matching engines, each driven by **4** bot types and ranked on a live leaderboard.
- Ran every untrusted submission in a **locked-down container** (seccomp, AppArmor, dropped capabilities, read-only non-root FS) and wrote a **shadow order book** that catches impossible fills regardless of execution order.

Cloud IDE 🌀 | React, TypeScript, FastAPI, Python, WebSockets

- Built a multi-user **browser IDE** with a VS Code editor (**Monaco**) and a live shell over WebSockets, plus sandboxed execution and a one-click full-stack runner.
- Structured a **multi-tenant** backend where a few shared servers back **thousands** of per-project databases across SQLite/PostgreSQL/MySQL/MongoDB, with remote-capable live preview.

RESEARCH EXPERIENCE

Undergraduate Researcher, CCNSB Lab, IIIT Hyderabad

Jan 2026 - Present

Advisor: Prof. Chittaranjan Hens | Complex Networks & Dynamical Systems

Hyderabad, India

- Developed Python and C++ simulators for flash-crash 🌀 and financial-contagion dynamics 🌀, with a generator that scales to **1,000,000-node** networks and recovers the exact known critical exponents.

TECHNICAL SKILLS

Languages: Python, Golang, C++, C, JavaScript, TypeScript, SQL, Bash/Shell

AI & Machine Learning: Deep Learning, Transformers & LLMs, LLM Inference & Serving, KV-Cache, Model Quantization (int4), Speculative Decoding, PyTorch

Frameworks & Libraries: React.js, Node.js, Express.js, FastAPI, Prisma ORM, Vite, Tailwind CSS, Socket.IO, Chart.js, NumPy, Pandas, Scikit-learn, OpenCV

Databases & Infrastructure: PostgreSQL, MongoDB, MySQL, SQLite, TimescaleDB, Redis, Kafka, Git, Docker, Kubernetes, gRPC/Protobuf, AWS, Linux

Core Competencies: Data Structures & Algorithms, Distributed Systems, System Design, High-Performance Computing (SIMD/AVX2), Concurrency, REST APIs, WebSockets

ACHIEVEMENTS & LEADERSHIP

INMO Qualifier (2022): Qualified for the Indian National Mathematical Olympiad (among the top ~300 nationally).

INMOTC (2022): Selected for INMO Training Camp at ICTS-TIFR, Bengaluru.

NSEA, Top 1% Karnataka: Top 1% statewide in the National Standard Examination in Astronomy (IAPT).

Math Olympiad Instructor, CFAL Mangaluru: Teach the four olympiad domains (algebra, geometry, combinatorics, number theory) for national olympiad preparation.

SGFI National Swimming: Represented Karnataka in Swimming at the School Games Federation of India Nationals.